

The role of epistemic beliefs in predicting ChatGPT adoption and avoidance in higher education

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ABSTRACT

AI-based tools such as ChatGPT are currently reshaping the landscape of higher education and significantly impacting student learning and academic achievement. With such powerful tools, individuals might get the impression of being guided by an omniscient epistemic authority that can be trusted absolutely and has the right answer to every question. However, little empirical research has yet addressed how these epistemic beliefs are associated with the adoption or avoidance of ChatGPT in higher education. Based on the theoretical frameworks of Hofer and Pintrich (1997) as well as Schommer (1990), we conducted a questionnaire study to assess epistemic beliefs about generative artificial intelligence and its adoption or avoidance. The sample consisted of 185 students (74.6% female) with an average age of 22.50 years from a German university. The questionnaire assessed the source and certainty of knowledge, the justification of knowledge claims, and the speed of knowledge acquisition. Results show that students' epistemic beliefs strongly predict their adoption of ChatGPT and moderately predict their avoidance of it. Epistemic beliefs thus emerge as an important factor in explaining learners' engagement with generative AI in higher education.

1. Introduction

In recent years, the use of artificial intelligence (AI) in education has increased enormously worldwide (Acosta-Enriquez et al., 2024; Sanabria-Navarro et al., 2023). In particular, the further development of Open AI's Chat Generative Pretrained Transformer (ChatGPT; OpenAI, 2023) has led to a veritable hype, making ChatGPT-4 the fastest-growing technology application in history (Milmo, 2023; Rajabi et al., 2024; Rudolph et al., 2023). Generative tools such as ChatGPT do not only produce output but can generate new, previously unseen information dependent on the data on which they were trained (García Peñalvo & Vázquez Ingelmo, 2023). With its ability to create extensive human-like output in a matter of seconds, the AI chatbot has gained particular significance in higher education (R. Wu & Yu, 2023). ChatGPT can explain complex topics from different subjects in simple terms, summarize texts, assist with exam preparation through interactive quizzes and study plans, and support the writing of term papers and essays. The quality is so advanced that it corresponds to a well above-average verbal IQ (Hickman et al., 2024), has the potential to pass graduate-level exams (Stribling et al., 2024), and gets teachers into trouble as they cannot distinguish between AI-generated essays and those written by students

(Fleckenstein et al., 2024).

With such a powerful tool as ChatGPT, individuals might get the impression of being guided by an omniscient epistemic authority (Cooper, 2023). Epistemic authorities can be a person, institution, or source that is considered authoritative for a particular field or issue (Cooper et al., 2024). It can be trusted absolutely and has the right answer to every question. Due to ChatGPT's extensive information base, its answers appear to be well-founded and justified (Wang et al., 2023). Knowledge seems to be acquired quickly through a one-time explanation of the topic (Lin, 2002; Schommer, 1990). Such views that individuals may hold about the nature of knowledge and knowledge acquisition with ChatGPT are known as epistemic beliefs. Schommer (1990) and later Hofer and Pintrich (1997) argued that epistemic beliefs consist of several independent but related dimensions such as the source and certainty of knowledge, the speed of knowledge acquisition, or the justification of knowledge claims.

There is ample reason to believe that the successful and meaningful use of AI in education depends on the ways in which learners' epistemic beliefs shape their engagement with AI-generated outputs (Richter & Schmid, 2010). However, little empirical research has yet addressed how epistemic beliefs specifically affect learners' use of and interactions

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with generative AI (Brändle et al., 2025; Urban et al., 2025). This article therefore aims to explore how epistemic beliefs are associated with the adoption or avoidance of ChatGPT in higher education.

2. Theoretical background

Research into individuals' epistemic beliefs about knowledge and knowledge acquisition in technology-supported learning environments began in the 1990s. This research activity specifically investigated epistemic beliefs about hypertext, a system of interconnected documents and digital information that allows users to navigate between content. Hypertext formed the technical basis for the development of the Internet which—with its vast amount of information—subsequently became the basis for generative AI. Therefore, we consider it important to briefly reflect on epistemic beliefs in relation to these important technological milestones.

2.1. Hypertext-related epistemic beliefs

Jacobson and Spiro (1995) were among the first to use a unidimensional questionnaire to demonstrate the impact of epistemic beliefs on student learning within a hypertext environment. Their findings indicated that more sophisticated epistemic beliefs facilitated learning in the non-linear structures characteristic for a hypertext system.

Building on theoretical work by Schommer (1990), Bendixen and Hartley (2003) used the multidimensional Epistemic Belief Inventory (Schraw et al., 2002) to explore how these beliefs influence learning with multimedia hypertexts. Their results show that beliefs in omniscient authorities and fixed abilities negatively affected knowledge acquisition, whereas a belief in quick learning positively influenced learning outcomes in the hypertext environment. Similarly, Bråten and Strømsø (2006) found that a belief in the speed of knowledge acquisition was positively associated with increased search activity on the Internet. However, this came at the cost of using deeper, comprehension-enhancing text-processing strategies.

2.2. Internet-related epistemic beliefs

Another important step forward was taken by Bråten et al. (2005) in relating epistemic beliefs to learning with Internet technologies. The Norwegian authors developed an Internet-Specific Epistemic Questionnaire (ISEQ) with 36 items based on the theoretical framework of personal epistemology by Hofer and Pintrich (1997). Students were asked to rate their courses on a ten-point scale in terms of whether the knowledge on the Internet is certain or tentative, simple or complex, a more important source than their own thinking, and whether knowledge claims on the Internet should be accepted or critically evaluated. Factor analysis revealed two distinct factors: General Internet Epistemology, which encompassed the certainty, simplicity, and source of knowledge on the Internet, and Justification for Knowing. These factors were better predictors of self-reported Internet search and communication activities than Internet self-efficacy beliefs (Bråten et al., 2005).

The ISEQ developed by Bråten et al. (2005) has been used in a whole series of further studies. The questionnaire provided important insights into the significance of epistemic beliefs for online academic help seeking (Cheng et al., 2013), the evaluation of and decisions about medical search results on the Internet (Kammerer et al., 2013), as well as self-regulated learning and academic information seeking on the Internet (Lee et al., 2014; Strømsø & Bråten, 2010). In addition, the questionnaire was further refined, and after revision a four-factor structure could be demonstrated (Chiu et al., 2013; Huang et al., 2023) in accordance with the dimensions proposed by Hofer and Pintrich (1997). It is worth noting that all four scales had a positive correlation with self-regulated learning; however, only the scale on justification for Internet-based knowing was found to predict negative behavior. Specifically, individuals who were less inclined to justify

knowledge claims found on the Internet were more likely to engage in academic dishonesty (Huang et al., 2023).

Finally, the epistemic beliefs concerning the justification of knowledge claims encountered on the Internet were further differentiated into three dimensions within the Internet-Specific Epistemic Justification Inventory (ISEJ; Kammerer et al., 2015; Bråten et al., 2019). The first scale, personal justification, describes how individuals attempt to reconcile information from the Internet with their personal knowledge and understanding. The other two scales address justification by consulting multiple sources and relying on expert authorities (Bråten et al., 2019). The importance of Internet-specific justification beliefs became evident in the clarification of controversial socio-scientific issues using the Internet (Kammerer et al., 2020) or in evaluating the credibility of multiple online documents (Hämäläinen et al., 2021; Bråten et al., 2022).

2.3. ChatGPT-related epistemic beliefs

Generative AI—in contrast to the Internet—is no longer merely a passive tool but increasingly functions as an active epistemic agent (J.-Y. Wu et al., 2025). This development facilitates novel forms of knowledge creation. Technology and users influence each other by jointly creating knowledge in a shared network (Scardamalia, 2002). Generative AI adapts to and responds to the input it receives from users. Conversely, users are also not passive recipients. Through their continuous engagement, they initiate and shape the learning processes and outcomes of generative AI.

Users' epistemic stances, which are closely tied to their justification of knowing (Bråten et al., 2014), play a critical role in this symbiotic interaction process (J.-Y. Wu et al., 2025). In an absolutist stance, no justification is considered necessary and AI-generated knowledge is accepted without question. In a multiplist stance, personally justified knowledge complements the knowledge of AI. In an evaluativist stance, AI-generated knowledge is critically examined, and users become additionally active by generating their own explanations and seeking corroborating evidence from multiple sources (Nussbaum et al., 2008). Within higher education, these dynamics underscore the need to examine how students perceive and engage with generative AI: whether they regard it as a neutral productivity aid or as an epistemic partner capable of shaping their knowledge practices.

Empirical research on the role of epistemic beliefs in the use of ChatGPT remains limited, with only two studies conducted in higher education to date. Brändle et al. (2025), using a questionnaire-based approach, found that students who frequently use AI chatbots tend to hold less sophisticated epistemic beliefs. Urban et al. (2025) developed a ChatGPT-specific questionnaire on epistemic beliefs based on Strømsø and Bråten (2010) with three scales and three items each: source and certainty of knowledge, structure of knowledge, and justification for knowing. In their study, students were asked to write a short report for a school newspaper based on two sources—one article written by an expert and another generated by ChatGPT. Notably, beliefs about the structure of knowledge, the perceived richness of information in ChatGPT, were associated with less use of information from the expert-written article. Students who strongly believed in the justification of ChatGPT-derived knowledge used more incorrect information from the ChatGPT-generated article. Overall, the current body of evidence is too limited to support broad conclusions about the impact of ChatGPT-specific epistemic beliefs in higher education.

2.4. ChatGPT usage and concerns

With the continued development of ChatGPT into a multifunctional tool and its applicability across various subject areas, research into its use in higher education has grown significantly (e.g., Al-Okaily et al., 2025; Habibi et al., 2023; Strzelecki, 2024). Baig and Yadegaridehkordi (2024) identified 57 quality-assured articles from 2023 and 2024 that

focus on the application of ChatGPT in educational contexts, including 14 studies specifically addressing its adoption and actual or intended use. An anonymous survey of more than 6300 students in Germany found that over two-thirds had used generative AI for academic purposes, with ChatGPT being by far the most widely used AI-based tool (von Garrel & Mayer, 2023).

Despite its clear advantages such as interactive and personalized learning, linguistic accuracy, user-friendliness, and enhanced productivity (Habibi et al., 2023), the usage of generative AI in higher education also presents notable challenges. These include diminished intellectual engagement, reduced critical thinking and independent problem-solving (Chan & Hu, 2023; Lo et al., 2024), overreliance on technology (Cai et al., 2024), and even a potential decline in creativity due to an AI-driven monoculture (Avraamidou, 2024). Moreover, ethical concerns have emerged, such as the use of fabricated data or sources, and the intentional circumvention of academic integrity guidelines for thesis writing (Farhi et al., 2023). It is therefore essential not only to examine the students' motivations for using ChatGPT in higher education, but also to seriously consider the technological and ethical concerns raised by those students who oppose its use (Farhi et al., 2023; Strzelecki, 2024).

2.5. Hypotheses

Since the number of studies on the relationship between epistemic beliefs and the adoption or avoidance of generative AI is limited and data is largely lacking, hypotheses can be formulated only on a general level. Brändle et al. (2025) found that individuals with naïve epistemic beliefs tend to perceive AI chatbots as comprehensive sources of knowledge and use them relatively often and uncritically for various purposes. Based on this, we propose the first hypothesis:

(1) AI-specific epistemic beliefs can explain the adoption of ChatGPT. Students with strong beliefs about the nature of knowledge and knowledge acquisition with ChatGPT will use it more often instrumentally and as an intelligent assistance tool for their work.

Regarding the avoidance of generative AI, the basis for hypotheses may be found in studies on Internet-triggered academic dishonesty (Akbulut et al., 2008). These include unethical behaviors such as plagiarism, data falsification, and the inappropriate use of online content and information. Huang et al. (2023) found that less sophisticated epistemic beliefs are associated with a higher likelihood of engaging in such undesired online behaviors. Based on this, we propose the second hypothesis:

(2) AI-specific epistemic beliefs can explain the avoidance of ChatGPT. Students with strong beliefs about the nature of knowledge and knowledge acquisition with ChatGPT are less inclined to express technological and ethical concerns.

3. Method

3.1. Sample

The sample for the study consisted of 185 students from a German university. Among the participants 138 identified as female and 46 as male, with one individual not specifying their gender. The average age of the study sample was 22.50 years ($SD = 3.38$). Participants had completed $M = 5.97$ semesters of study ($SD = 3.79$) across a range of disciplines, including economics, computer science, law, journalism, media studies, political science, European studies, cultural economics, developmental studies, sociology, education, and various teacher training programs. As this was a broad student sample, no particular focus on specific subjects could be found.

3.2. Materials

The questionnaire designed to assess epistemic beliefs about

ChatGPT comprised four scales and was informed by the multidimensional frameworks of Hofer and Pintrich (1997) as well as Schommer (1990). In addition to beliefs about the source, certainty, and justification of ChatGPT's knowledge, the speed of knowledge acquisition using ChatGPT was identified as a relevant dimension for evaluation. To assess participants' agreement with each item, an 11-point Likert scale was employed. Responses were recorded in 10% increments, ranging from 0% = *strongly disagree* (coded as 1) to 100% = *strongly agree* (coded as 11).

3.2.1. Source of knowledge

Users of ChatGPT may risk perceiving this form of generative artificial intelligence as an ultimate epistemic authority (Cooper, 2023), accepting its outputs without sufficient scrutiny. High scores on this scale indicate a relatively uncritical stance toward the information provided by ChatGPT, reflecting a tendency to trust its knowledge claims. The scale included five items, adapted from Conley et al. (2004), and demonstrated good reliability (Cronbach's $\alpha = 0.83$). A sample item is: "Whatever knowledge ChatGPT imparts is correct."

3.2.2. Certainty of knowledge

ChatGPT users may vary in their perceptions of whether the knowledge it generates is fixed and absolute or instead tentative and subject to change (Urban et al., 2025). This central dimension of epistemological belief systems was assessed with five items, also based on Conley et al. (2004), and showed strong reliability ($\alpha = 0.85$). High scores on this scale reflect the belief that AI-generated knowledge is certain and not open to revision. A sample item is: "ChatGPT already knows most things; there is not much it does not know yet."

3.2.3. Justification of knowledge claims

ChatGPT users may hold different beliefs about whether and how to justify knowledge claims. Some learners may seek to substantiate knowledge in a process of symbiotic agency with the technological tool (J.-Y. Wu et al., 2025). The scale justification of knowledge claims consisted of four items, developed with reference to Conley et al. (2004), and exhibited moderate but acceptable reliability ($\alpha = 0.65$). High scores on this scale suggest a tendency to justify ideas and concepts with the help of generative AI. This understanding of the justification of knowledge claims differs from earlier Internet-specific conceptions (Bråten et al., 2019) in that generative AI is now granted an active role in the justification process. A sample item is: "In ChatGPT, there may be several ways to validate views and ideas."

3.2.4. Speed of knowledge acquisition

Some users may perceive ChatGPT as a tool that accelerates the learning process by providing immediate access to relevant information and enabling targeted inquiries. Drawing on Schommer (1990), four items were constructed to capture beliefs about fast, machine-assisted learning. The scale demonstrated acceptable reliability ($\alpha = 0.70$). High scores on this scale reflect the belief that ChatGPT facilitates rapid learning and comprehension. A sample item for quick learning is: "With ChatGPT, you can understand everything immediately."

Schommer (1990) held the view that epistemic beliefs do not develop in a uniform manner, as suggested by Perry Jr (1970), but rather represent separate, relatively independent dimensions. In line with Schommer's multidimensional perspective, a model with four independent but related factors was tested for this study. The confirmatory factor analysis with SPSS Amos 29 (Arbuckle, 2022) demonstrated a good fit to the data for the four-factor model, $\chi^2 = 208.48$, $df = 119$, $\chi^2/df = 1.75$, $p < .05$, CFI = 0.94, RMSEA = 0.06, 90% CI [0.049; 0.078], SRMR = 0.07 (Hu & Bentler, 1999). An alternative model with three factors, in which the source and certainty of knowledge are combined into one factor (see Strømso & Bråten, 2010), yielded a less satisfactory fit, $\chi^2 = 251.77$, $df = 132$, $\chi^2/df = 1.91$, $p < .05$, CFI = 0.91, RMSEA = 0.07, 90% CI [0.057; 0.083], SRMR = 0.06. A Chi-square difference test

(Kline, 2011) indicated that the four-factorial model fitted the data significantly better than the three-factorial model ($\chi^2 = 43.29$, $df = 13$, $p < .05$).

A questionnaire developed by Farhi et al. (2023) was adapted to explore students' views and concerns regarding the use of generative artificial intelligence. The original instrument comprises four scales: ChatGPT Usage, Students' Views, Students' Concerns, and Perceived Ethics. For the present study, the questionnaire was translated and administered to students to assess their attitudes toward generative AI.

As in the original study (Farhi et al., 2023), all items were rated on a five-point Likert scale with the following response options: 1 = *strongly disagree*, 2 = *somewhat disagree*, 3 = *neutral*, 4 = *somewhat agree*, and 5 = *strongly agree*. The four hypothesized factors were tested using confirmatory factor analysis. The model demonstrated an acceptable fit to the data, $\chi^2 = 320.53$, $df = 187$, $\chi^2/df = 1.71$, $p < .05$, CFI = 0.90, RMSEA = 0.06, 90% CI [0.051; 0.074], SRMR = 0.07 (Hu & Bentler, 1999). The factor names were slightly modified to better reflect the specific content of each scale.

3.2.5. Instrumental usage

This scale captures the use of ChatGPT as an advanced tool for solving routine academic tasks, such as answering questions or locating information sources. The five items of the scale demonstrated acceptable reliability (Cronbach's $\alpha = 0.69$). A sample item is: "I use ChatGPT because it offers me an effective solution."

3.2.6. Intelligent assistance

This scale reflects the use of ChatGPT for more complex tasks, such as writing, translation, and understanding user input to provide contextually appropriate responses. The six items showed acceptable reliability ($\alpha = 0.73$). A sample item is: "I use ChatGPT because it provides me with good writing suggestions."

3.2.7. Technological concerns

This scale addresses apprehensions about the technological impact of generative AI, including fears of diminished critical thinking or creative writing skills and the risk of overreliance on the technology. The five items demonstrated good reliability ($\alpha = 0.75$). A sample item is: "I do not use ChatGPT because it has a negative effect on critical thinking."

3.2.8. Ethical concerns

This scale captures ethical issues associated with the use of ChatGPT in educational settings, such as presenting AI-generated work as one's own or relying on unverified sources. The six items showed acceptable reliability ($\alpha = 0.72$). A sample item is: "I do not use ChatGPT because it can lead me into ethical dilemmas."

3.3. Procedure

The study was conducted as part of an experiment investigating learning with the use of analog and digital media. A trained experimenter instructed and supervised groups of 2 to 9 participants during the learning phase. Participation was anonymous and voluntary, and students received €10 compensation for the approximately one-hour session. The data reported here was collected via computer at the beginning of the study.

Participants were informed in writing of their right to withdraw from the study at any time and to terminate their participation without providing a reason. However, no participant made use of this option, meaning that there are no missing data in the data set. The chair of the university's ethics committee was consulted in advance and fully informed about the planned project. No ethical or safety-related concerns were raised regarding the study's implementation.

3.4. Statistical analyses

Correlation analyses and multiple linear regressions were conducted to test the hypotheses. These types of analyses are advantageous in that they can reliably detect significant effects even with relatively small sample sizes. A power analysis using G*Power (Faul et al., 2009) for a linear multiple regression with six predictors, a medium effect size of $f^2 = 0.15$, an alpha level of 0.05, and a power ($1-\beta$) of 0.80, indicated a minimum sample size of 98 participants. At this effect size, the predictor set would account for 15% of the variance in the dependent variable (Memon et al., 2020).

Hierarchical stepwise linear regression analyses were used to identify which epistemic beliefs were most strongly associated with the acceptance or rejection of ChatGPT. This method combines hierarchical regression, where predictors are entered in blocks based on theoretical rationale, with stepwise regression, which selects predictors based on statistical significance. In the first block, background variables such as gender and number of semesters were included to predict ChatGPT adoption or avoidance. In the second block, epistemic belief variables were added sequentially based on their correlational strength with the dependent variable, until no further variables contributed significantly to the explained variance. Significant predictors in the regression analyses are then transferred to a path analysis model with manifest variables to get a detailed understanding of the relationships between epistemic beliefs and ChatGPT adoption or avoidance.

4. Results

4.1. Descriptive statistics

Table 1 presents the means and standard deviations of the study variables. The mean values of epistemic beliefs were tested using one-sample *t*-tests against the theoretical midpoint ($\mu = 6$) of the 11-point Likert scale. Results indicated that students rated the source ($t = -19.25$, $df = 184$, $p < .001$, $d = 1.42$) and certainty ($t = -19.16$, $df = 184$, $p < .001$, $d = 1.41$) of ChatGPT's knowledge below the midpoint. In contrast, the justification of knowledge claims ($t = -7.66$, $df = 184$, $p < .001$, $d = 0.56$) and the speed of knowledge acquisition ($t = 6.99$, $df = 184$, $p < .001$, $d = 0.51$) were rated slightly above average. Overall, students' attitudes toward the knowledge generated by ChatGPT appear to be somewhat critical.

However, a different picture emerges regarding the acceptance of and concerns about using the chatbot. One-sample *t*-tests against the theoretical midpoint ($\mu = 3$) on a five-point Likert scale revealed notable findings related to the use and rejection of ChatGPT. Specifically, mean values for instrumental usage ($t = 7.48$, $df = 184$, $p < .001$, $d = 0.55$) and intelligent assistance ($t = 4.88$, $df = 184$, $p < .001$, $d = 0.36$) were above the midpoint, while technological concerns ($t = -1.84$, $df = 184$, ns , $d = 0.13$) did not differ significantly from the midpoint, and ethical concerns ($t = -14.46$, $df = 184$, $p < .001$, $d = 1.06$) were below the midpoint. These results suggest that attitudes toward the use of ChatGPT are generally positive and ethical concerns are considered secondary.

4.2. Correlational analyses

Table 1 also presents the intercorrelations among the study variables. To address the hypotheses, the relationships between epistemic beliefs and both the adoption and avoidance of ChatGPT needs to be considered.

Consistent with the first hypothesis, all epistemic beliefs about generative AI show clear and significant positive correlations with both the instrumental usage of ChatGPT and its perceived intelligent assistance. A strong belief in the knowledge provided by ChatGPT, as well as in its role in knowledge acquisition, is associated with greater usage of the tool.

In line with the second hypothesis, epistemic beliefs also appear to

Table 1

Descriptive statistics and intercorrelations of the study variables.

Variable	<i>M</i>	<i>SD</i>	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) Source of knowledge	3.76	1.58	—									
(2) Certainty of knowledge	3.62	1.69	0.80**	—								
(3) Justification of knowledge claims	6.82	1.46	0.44**	0.39**	—							
(4) Speed of knowledge acquisition	6.84	1.64	0.57**	0.49**	0.52**	—						
(5) Instrumental usage	3.46	0.84	0.35**	0.27**	0.55**	0.49**	—					
(6) Intelligent assistance	3.29	0.81	0.40**	0.27**	0.42**	0.57**	0.54**	—				
(7) Technological concerns	2.88	0.89	−0.19**	−0.16*	−0.41**	−0.13	−0.29**	−0.30**	—			
(8) Ethical concerns	2.27	0.69	−0.17**	−0.14	−0.37**	−0.20**	−0.33**	−0.33**	0.70**	—		
(9) Gender ^a	1.26	0.44	−0.01	−0.06	−0.02	−0.11	−0.07	−0.09	−0.05	−0.05	—	
(10) Age	22.50	3.38	−0.22**	−0.20**	−0.04	−0.16*	0.05	−0.00	0.00	0.06	0.16*	—
(11) Semester	5.97	3.79	−0.15*	−0.12	0.04	−0.12	−0.06	−0.04	−0.17*	−0.16*	0.07	0.50**

^a Female = 1, male = 2.* $p < .05$.** $p < .01$.

explain the avoidance of ChatGPT. Epistemic beliefs regarding the source, certainty, and justification of knowledge correlate significantly and negatively with technological concerns. Participants who are more confident in ChatGPT's epistemic reliability tend to express less apprehension about the technological impact of generative AI on individuals. Furthermore, beliefs about the source and justification of knowledge, along with expectations regarding the speed of knowledge acquisition, correlate significantly and negatively with ethical concerns. Those who trust in ChatGPT's knowledge and anticipate quick learning through AI tend to report fewer ethical reservations about using the chatbot.

4.3. Regression analyses

Linear multiple regression analyses were conducted to examine the statistical significance of various epistemic beliefs in predicting both the adoption and avoidance of ChatGPT. Specifically, four hierarchical stepwise linear regressions were performed. Box plots were used to identify systematic outliers, but none were found. A test for homoscedasticity, based on a scatter plot of standardized residuals versus predicted values, revealed no systematic patterns or funnel-shaped distributions. To ensure that all predictors are stable and interpretable, tolerance and variance inflation factor were checked. All values were in the normal range with tolerance values not less than 0.64 and the variance inflation factor never greater than 1.56. Gender and number of semesters were included as control variables to ensure that the effects of epistemic beliefs on ChatGPT adoption and avoidance were not confounded by third variables. Age was excluded as a control variable

due to its strong correlation with the number of semesters (see Table 1), in order to avoid multicollinearity among the predictor variables.

The regression results presented in Tables 2 and 3 support the first hypothesis that epistemic beliefs can predict the adoption of ChatGPT. In both models, justification of knowledge claims and speed of knowledge acquisition emerged as significant predictors. Justification of knowledge claims was the strongest predictor of the instrumental usage of ChatGPT, while speed of knowledge acquisition best predicted the perception of ChatGPT as an intelligent assistant. The explained variance in both models is relatively high—accounting for approximately one-third of the total variance—and qualifies as a large effect size according to Cohen's (2013) guidelines.

The regression results shown in Tables 4 and 5 support the second hypothesis that epistemic beliefs can also predict the avoidance of ChatGPT. In both cases, justification of knowledge claims significantly contributed to explaining students' technological and ethical concerns that deter them from using ChatGPT. The less justified ChatGPT's knowledge, the greater the concerns expressed across various domains. In addition, a higher number of semesters was associated with fewer technological and ethical concerns. According to Cohen's (2013) guidelines, the explained variance in both cases represents a medium effect size.

A path analysis model with manifest variables was created to summarize the significant results of the regression analyses. The model in Fig. 1 shows a good fit to the data, $\chi^2 = 3.78$, $df = 2$, $\chi^2/df = 1.89$, $p < .05$, CFI = 0.99, RMSEA = 0.069, 90% CI [0.000; 0.177], SRMR = 0.02. Speed of knowledge acquisition and justification of knowledge claims

Table 2

Hierarchical stepwise regression results for instrumental usage.

Variable	<i>B</i>	95% CI for <i>B</i>		<i>SE B</i>	β	<i>R</i> ²	ΔR^2
		<i>LL</i>	<i>UL</i>				
Step 1						0.007	0.007
Constant	3.66***	3.28	4.05	0.20			
Gender ^a	−0.11	−0.36	0.15	0.13	−0.06		
Semester	−0.01	−0.04	0.02	0.02	−0.05		
Step 2						0.315	0.308***
Constant	1.49***	0.92	2.06	0.29			
Gender ^a	−0.08	−0.30	0.13	0.11	−0.05		
Semester	−0.02	−0.04	0.01	0.01	−0.07		
Justification	0.32***	0.25	0.39	0.04	0.56		
Step 3						0.363	0.049***
Constant	1.01***	0.40	1.62	0.31			
Gender ^a	−0.04	−0.25	0.16	0.11	−0.02		
Semester	−0.01	−0.04	0.02	0.01	−0.04		
Justification	0.24***	0.16	0.32	0.04	0.42		
Speed	0.14***	0.06	0.21	0.04	0.26		

Note. *N* = 184. CI = confidence interval; LL = lower limit; UL = upper limit.

^a Female = 1, male = 2.*** $p < .001$.

Table 3
Hierarchical stepwise regression results for intelligent assistance.

Variable	<i>B</i>	95% CI for <i>B</i>		<i>SE B</i>	β	<i>R</i> ²	ΔR^2
		<i>LL</i>	<i>UL</i>				
Step 1						0.009	0.009
Constant	3.52***	3.14	3.89	0.19			
Gender ^a	−0.14	−0.39	0.11	0.13	−0.08		
Semester	−0.01	−0.04	0.02	0.02	−0.04		
Step 2						0.326	0.317***
Constant	1.38***	0.84	1.93	0.28			
Gender ^a	−0.04	−0.25	0.16	0.10	−0.02		
Semester	0.01	−0.02	0.03	0.01	0.03		
Speed	0.28***	0.22	0.34	0.03	0.57		
Step 3						0.346	0.020*
Constant	1.09***	0.49	1.69	0.30			
Gender ^a	−0.05	−0.25	0.15	0.10	−0.03		
Semester	0.00	−0.02	0.03	0.01	0.01		
Speed	0.24***	0.17	0.31	0.04	0.48		
Justification	0.09*	0.01	0.17	0.04	0.17		

Note. *N* = 184. *CI* = confidence interval; *LL* = lower limit; *UL* = upper limit.

^a Female = 1, male = 2.

* *p* < .05.

*** *p* < .001.

Table 4
Hierarchical stepwise regression results for technological concerns.

Variable	<i>B</i>	95% CI for <i>B</i>		<i>SE B</i>	β	<i>R</i> ²	ΔR^2
		<i>LL</i>	<i>UL</i>				
Step 1						0.032	0.032
Constant	3.22***	2.81	3.62	0.20			
Gender ^a	−0.08	−0.34	0.19	0.14	−0.04		
Semester	−0.04*	−0.07	−0.01	0.02	−0.17		
Step 2						0.192	0.160***
Constant	4.88***	4.22	5.54	0.33			
Gender ^a	−0.10	−0.34	−0.15	0.12	−0.05		
Semester	−0.04*	−0.07	−0.01	0.02	−0.16		
Justification	−0.24***	−0.32	−0.16	0.04	−0.40		

Note. *N* = 184. *CI* = confidence interval; *LL* = lower limit; *UL* = upper limit.

^a Female = 1, male = 2.

* *p* < .05.

*** *p* < .001.

Table 5
Hierarchical stepwise regression results for ethical concerns.

Variable	<i>B</i>	95% CI for <i>B</i>		<i>SE B</i>	β	<i>R</i> ²	ΔR^2
		<i>LL</i>	<i>UL</i>				
Step 1						0.026	0.026
Constant	2.50***	2.19	2.82	0.16			
Gender ^a	−0.06	−0.26	0.15	0.11	−0.04		
Semester	−0.03*	−0.05	−0.00	0.01	−0.15		
Step 2						0.162	0.136***
Constant	3.69***	3.17	4.21	0.27			
Gender ^a	−0.07	−0.26	0.13	0.10	−0.05		
Semester	−0.03*	−0.05	−0.00	0.01	−0.14		
Justification	−0.17***	−0.24	−0.11	0.03	−0.37		

Note. *N* = 184. *CI* = confidence interval; *LL* = lower limit; *UL* = upper limit.

^a Female = 1, male = 2.

* *p* < .05.

*** *p* < .001.

are significantly related to the dependent variables measuring the acceptance and avoidance of ChatGPT.

5. Discussion

AI-based tools such as ChatGPT are currently reshaping the landscape of higher education and significantly impacting student learning

and academic achievement (e.g., Liu et al., 2025; von Garrel & Mayer, 2023). Students' epistemic cognitions regarding generative artificial intelligence play a decisive role in whether they adopt or reject such supportive tools. The present study with students at a German university demonstrated that strong beliefs about knowledge and knowledge acquisition with ChatGPT promote its use in academic settings. In contrast, perceiving ChatGPT as having low epistemic value was

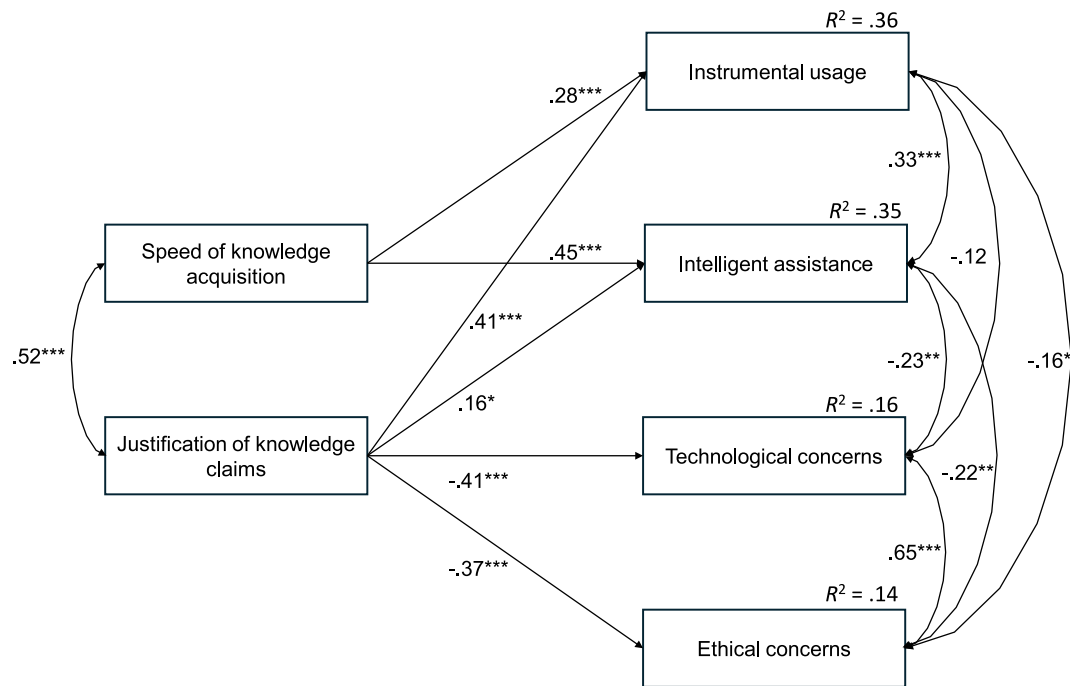


Fig. 1. Path analysis model of associations between epistemic beliefs and CHATGPT adoption and avoidance tendencies.

Note. The path analysis model summarizes the statistically significant results of the four hierarchical stepwise linear regression analyses. Coefficients presented are standardized linear regression coefficients.

* $p < .05$. ** $p < .01$. *** $p < .001$.

associated with increased technological and ethical concerns and with restricted use of the technology.

5.1. Interpretation of the findings

Our analyses indicate that German students are relatively skeptical about the source and certainty of ChatGPT's knowledge. They are aware that ChatGPT can fabricate references to seemingly scientific articles and that AI-generated responses are not always correct (Caspi & Karp, 2024). Students know that the reliability and credibility of the data are not yet sufficient for the chatbot to be used in higher education without reservation. The analyses further reveal that attitudes toward the use of ChatGPT are generally positive, and students' ethical concerns not to use it are relatively low. The benefits of employing this powerful tool appear to outweigh the drawbacks associated with its potential misuse and the ethical dilemmas it may pose. This suggests that students hold rather multiplist beliefs on generative AI: they know about the lack of credibility but still use ChatGPT as a supposedly reliable source of knowledge.

The current study pursued two hypotheses to investigate the relationship between epistemic beliefs and both the adoption and avoidance of ChatGPT. Supporting the first hypothesis, the findings indicate that strong, absolutist beliefs about knowledge and knowledge acquisition with ChatGPT promote its use. Epistemic cognitions moderately correlate with ChatGPT's instrumental usage and its perceived intelligent assistance. Regression analyses revealed that, among the four epistemic dimensions, justification of knowledge claims and speed of knowledge acquisition were particularly strong predictors. Students who perceive ChatGPT as offering diverse justification of knowledge claims and expect to learn quickly with the technology use the chatbot more instrumentally and as a means of intelligent learning support. These results align with the findings of Brändle et al. (2025). However, the direction of causality remains unclear: Do positive attitudes toward ChatGPT lead to greater use, or does heavy use of ChatGPT lead to a more positive attitude toward the medium? Only more experimental and longitudinal research can provide definitive insights here (Brady et al.,

2023).

Consistent with the second hypothesis, the results show that greater skepticism toward knowledge and knowledge acquisition with ChatGPT is associated with more technological and ethical concerns, ultimately limiting its use. This applies to almost all epistemic dimensions, but the justification of knowledge claims is a significant negative predictor of concerns about generative AI. These findings are in line with Huang et al. (2023) who found that sound justification and critical evaluation of online information correlate with lower instances of academic misconduct. Applied to our study, beliefs in poorly justified knowledge claims provided by ChatGPT appear to discourage its usage and promote more ethical engagement with the technology.

5.2. Limitations of the study

While this study focused on epistemic beliefs specifically related to the use of ChatGPT, it is important to acknowledge that epistemic beliefs are largely domain-specific (Muis et al., 2006), and students' views on the nature of knowledge and knowledge acquisition can vary significantly depending on their field of study (Urhahne & Kremer, 2023). Future research should build on the present findings by conducting comparative studies across predefined subjects or academic disciplines, which would provide more nuanced insights into how epistemic beliefs shape interactions with ChatGPT in different knowledge domains.

Hofer (2001) suggested that the concept of quick learning, or the speed of knowledge acquisition, may reflect a metacognitive dimension rather than a traditional epistemic belief. This observation relates to Schommer's (1990) conceptualization of epistemic beliefs, which underlies the notion of quick learning and informed the present study. Importantly, however, our findings underscore that the belief in quick learning is a meaningful predictor of ChatGPT acceptance. Whether conceptualized as an epistemic or metacognitive construct, students' perception that ChatGPT facilitates quick learning emerges as a key motivator for its use.

Measuring epistemic beliefs through questionnaires presents well-

known challenges, such as item abstraction and potential variability in participants' interpretations (Clarebout et al., 2001). In the current study, the reliability of the instruments ranged from acceptable to satisfactory, supporting the robustness of the findings. Future studies could benefit from refined measurement tools, which would allow for even more precise and nuanced assessment of epistemic and meta-cognitive beliefs in the context of AI-supported learning. This applies especially to the scale on the justification of knowledge claims which should be expanded to better reflect the epistemic belief of co-constructed knowledge between users and generative AI.

5.3. Implications and outlook

Future research on ChatGPT-specific epistemic beliefs should adopt a broader scope and complement subjective data with objective measures. This is particularly relevant for examining interaction behavior and knowledge acquisition in the context of chatbot use.

Generative artificial intelligence enables new forms of interactive engagement. Depending on learners' questions and prompting strategies, the algorithm generates responses that are tailored as closely as possible to the given context. This opens the possibility of investigating whether epistemic beliefs influence both the quality and quantity of interactions with the chatbot. For example, learners with an absolutist stance may quickly accept the generated response without further inquiry. In contrast, students with an evaluativist stance are more likely to critically examine the response and engage in deeper interaction by posing follow-up or challenging questions (J.-Y. Wu et al., 2025). This may also materialize in more elaborate ways of interacting with the chatbot such as assigning it a specific role (e.g., tutor, debate opponent, or editor), prompting it to justify its reasoning, or request multiple alternative explanations to compare and evaluate the information provided.

Such sophisticated interaction patterns are likely to influence the effectiveness of technology-supported knowledge acquisition. Learners with an absolutist stance, who engage less critically, may acquire less new knowledge using generative AI than those with a multiplist or evaluativist stance (J.-Y. Wu et al., 2025). This hypothesis should be tested in future studies focusing on learning processes and usage behavior involving generative AI. Such research could investigate not only the quantity of knowledge gained, but also the depth of understanding, the ability to critically evaluate AI-generated information, and the strategies learners employ to integrate AI outputs into their broader learning goals. Additionally, examining how these interaction patterns vary across domains or task types would provide valuable insights into tailoring AI-supported educational interventions to different epistemic profiles. Future research could explore how these interactions are associated with learning outcomes across different disciplines and tasks, providing further insight into the role of epistemic beliefs in AI-supported education.

ChatGPT is not free from biases and stereotypes that take influence on users' beliefs and attitudes (Yuan et al., 2025). It is therefore important to question the epistemic status of generative AI. Cooper et al. (2024) give some practical suggestions for teaching how ChatGPT can be perceived more as a dialogue partner and less as an epistemic authority providing absolute truths. Comparing the output with those of other large language models can reveal enormous differences, or the teacher can use collaborative prompting to stimulate a discussion with students about the responsible use of generative AI.

5.4. Conclusion

For about three decades, empirically oriented research has examined the relationship between epistemic beliefs and the use of educational technologies (Jacobson & Spiro, 1995). With the emergence of AI chatbots such as ChatGPT, a new epistemic dimension has opened: learners' epistemic beliefs not only shape their interactions with

computer technology, but generative AI also actively influences learners by offering prompts and presenting itself as a seemingly omniscient epistemic authority (Cooper, 2023). Within this intriguing dynamic, conceptions of the nature of knowledge and knowledge acquisition are being redefined.

This questionnaire-based study represents one of the first steps toward investigating the epistemic status of generative AI using a multi-dimensional approach (Hofer & Pintrich, 1997; Schommer, 1990). Our results show that students' epistemic beliefs strongly predict their adoption of ChatGPT and moderately predict their avoidance of it. These findings highlight the importance of epistemic beliefs in shaping learners' engagement with generative AI in higher education.

CRedit authorship contribution statement

Detlef Urhahne: Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Laura Fink:** Writing – review & editing, Project administration, Investigation, Formal analysis, Data curation. **Lars Dietrich:** Writing – original draft, Visualization, Validation. **Kerstin Kremer:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

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Appendix A. ChatGPT-Related epistemic beliefs

Source of Knowledge

1. You have to trust what ChatGPT disseminates. [Man muss darauf vertrauen, was ChatGPT verbreitet.]
2. You have to believe in the knowledge ChatGPT offers. [Man muss daran glauben, was ChatGPT an Wissen anbietet.]
3. Whatever knowledge ChatGPT imparts is correct. [Was auch immer ChatGPT an Wissen vermittelt, ist korrekt.]
4. ChatGPT knows exactly what is true. [ChatGPT weiß ganz genau, was wahr ist.]
5. ChatGPT has the right solution for every question. [Für alle Fragen hat ChatGPT die richtige Lösung parat.]

Certainty of Knowledge

6. ChatGPT already knows most things; there is not much it does not know yet. [ChatGPT weiß eigentlich schon das meiste; es gibt nicht mehr viel, was es noch nicht weiß.]
7. ChatGPT's knowledge is up to date. [Das Wissen von ChatGPT ist topaktuell.]
8. You can trust what you read in ChatGPT. [Die Dinge, die man in ChatGPT liest, kann man guten Gewissens glauben.]
9. The best thing about ChatGPT is that it has exactly one correct answer for every question. [Das Beste an ChatGPT ist, dass es für alle Fragen genau eine richtige Antwort bereithält.]
10. When you read something in ChatGPT, you can be sure it is true. [Wenn man in ChatGPT etwas liest, kann man sicher sein, dass es wahr ist.]

Justification of Knowledge Claims

11. Consulting ChatGPT is a good way to find out whether a piece of information is really true. [ChatGPT zu befragen, ist ein guter Weg, um herauszufinden, ob eine Erkenntnis wirklich wahr ist.]

12. In ChatGPT, there may be several ways to validate views and ideas. [In ChatGPT kann es mehrere Möglichkeiten geben, um Anschauungen und Vorstellungen zu überprüfen.]
13. In ChatGPT, knowledge can develop from your own questions and efforts. [In ChatGPT kann sich Wissen aus den eigenen Fragen und Bemühungen heraus entwickeln.]
14. ChatGPT can be creative by combining different knowledge bases. [ChatGPT kann kreativ sein, indem es verschiedene Wissensbestände miteinander verbindet.]

Speed of Knowledge Acquisition

15. With ChatGPT, you can understand everything immediately. [Mit ChatGPT kann man alles sofort verstehen.]
16. With ChatGPT, you can learn content quickly. [Mit ChatGPT kann man Inhalte schnell lernen.]
17. If I do not understand something, I ask ChatGPT. [Wenn ich etwas nicht verstehe, frage ich ChatGPT.]
18. When ChatGPT tells me something, it makes immediate sense to me. [Wenn mir ChatGPT etwas mitteilt, macht es für mich unmittelbar Sinn.]

Data availability

Data will be made available on request.

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